

Mapterhorn Terrain and Imagery

State of the Map Paris, 28.08.2026

Oliver Wipfli

About Me

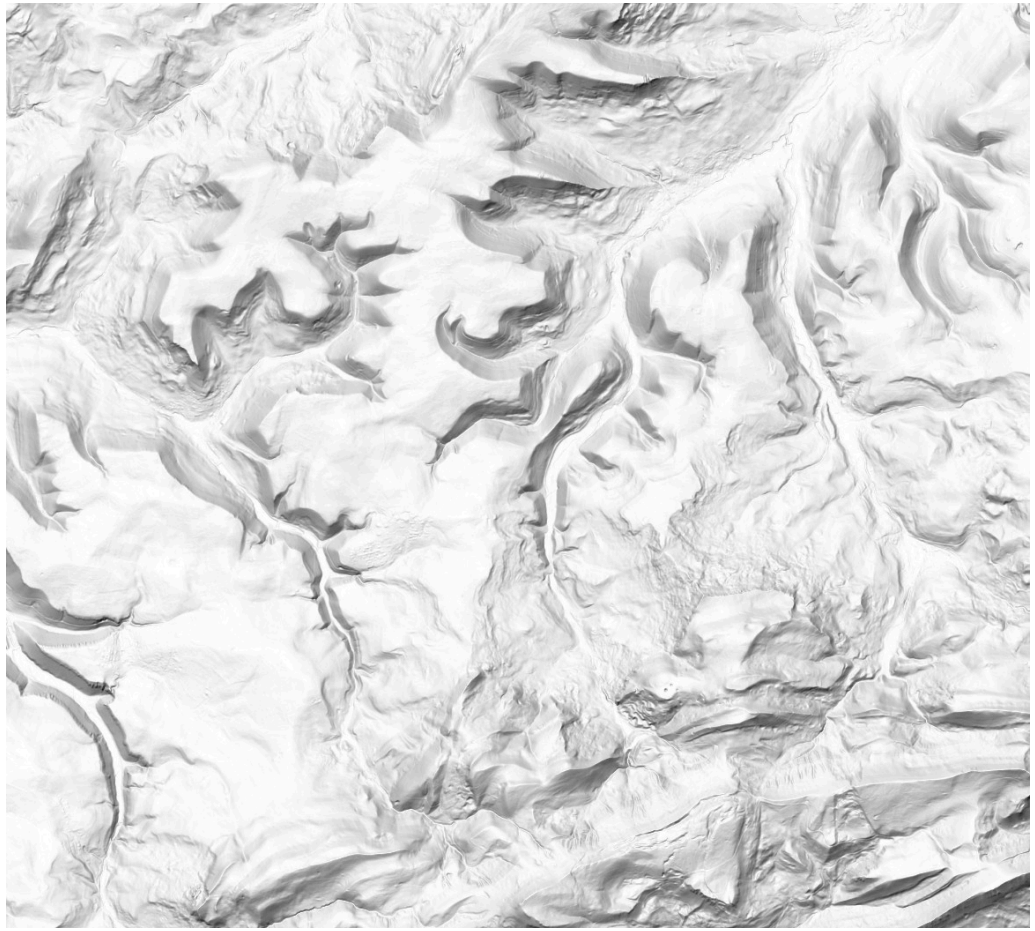
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- Consultant in Maps and Location Technology
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- GitHub: https://github.com/wipfli
- LinkedIn: oliver-wipfli-562258210



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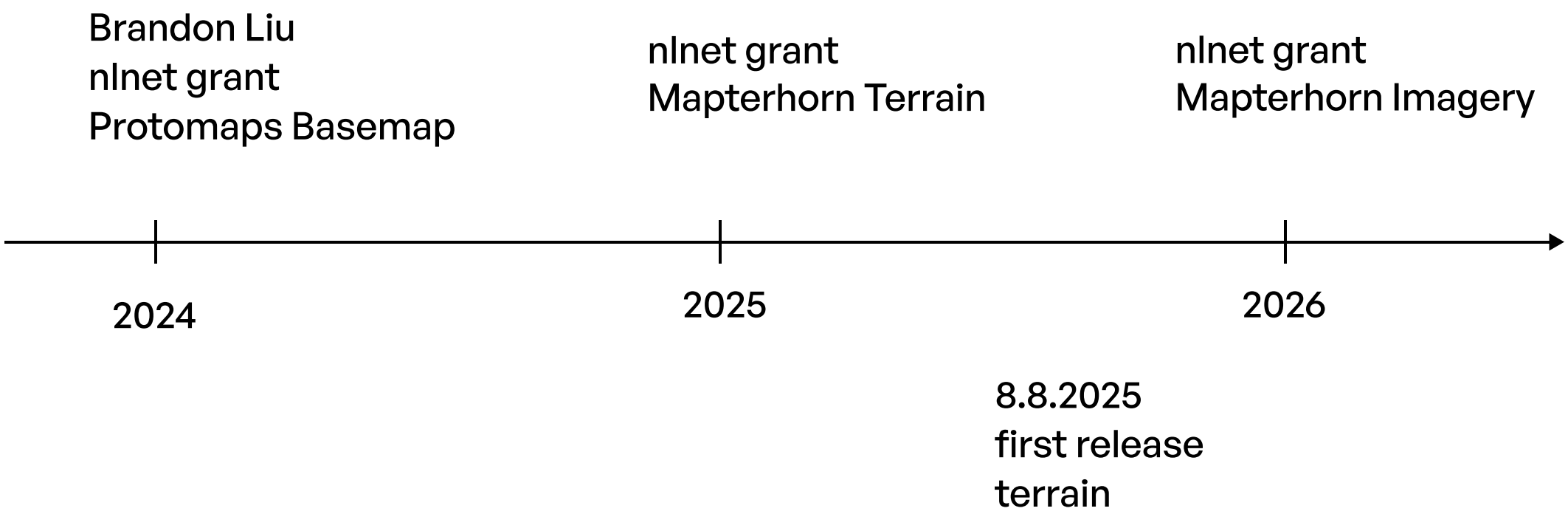
About Mapterhorn

- Aggregation of open terrain data
- Global low-resolution coverage, local high-resolution lidar
- For hillshading, contour lines, routing, u.s.w.
- Soon also aerial imagery



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History



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Outline

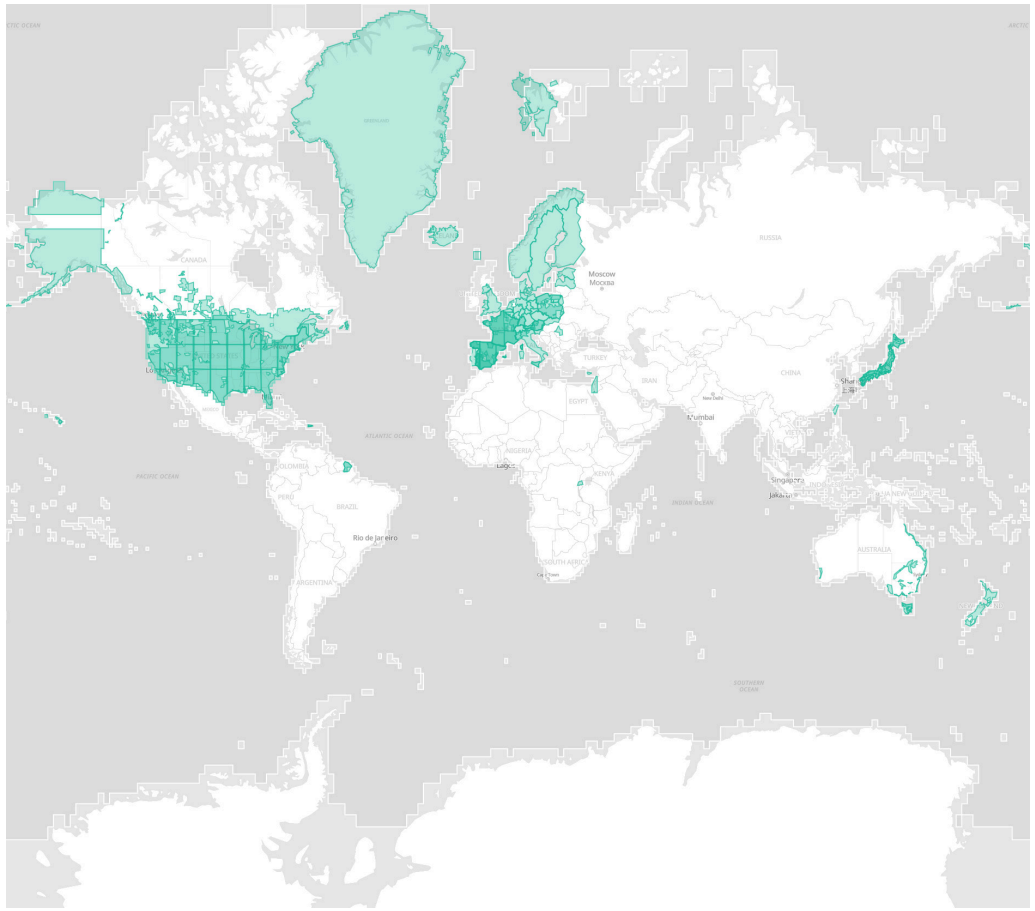
- **Input**
- Aggregation
- Output and Usage
- Sponsors
- Imagery
- Contributing



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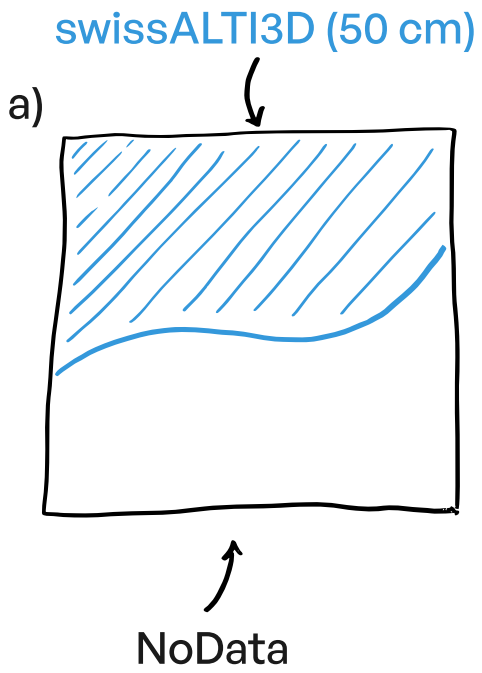
Input

- Global: Copernicus 30 m resolution
- Local: High-resolution Lidar
- 70 data producers
- 18 TiB total input data, shared as tarballs
- mapterhorn.com/attribution
- License CC-BY "or less"



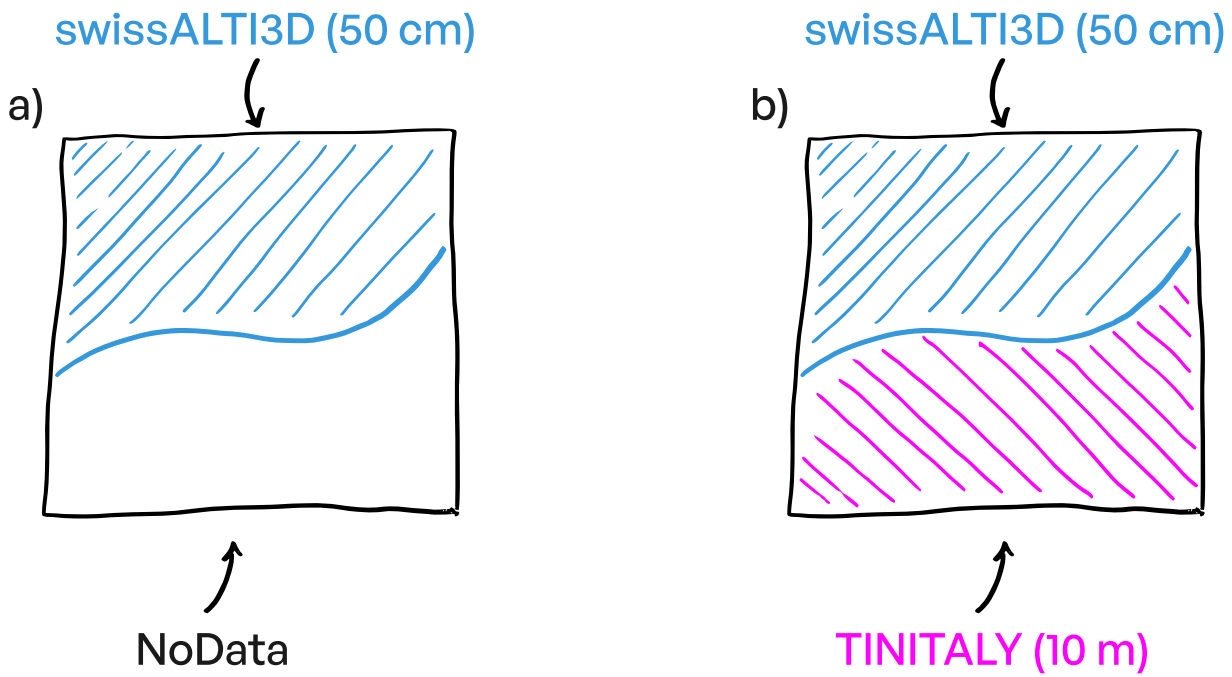
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Aggregation



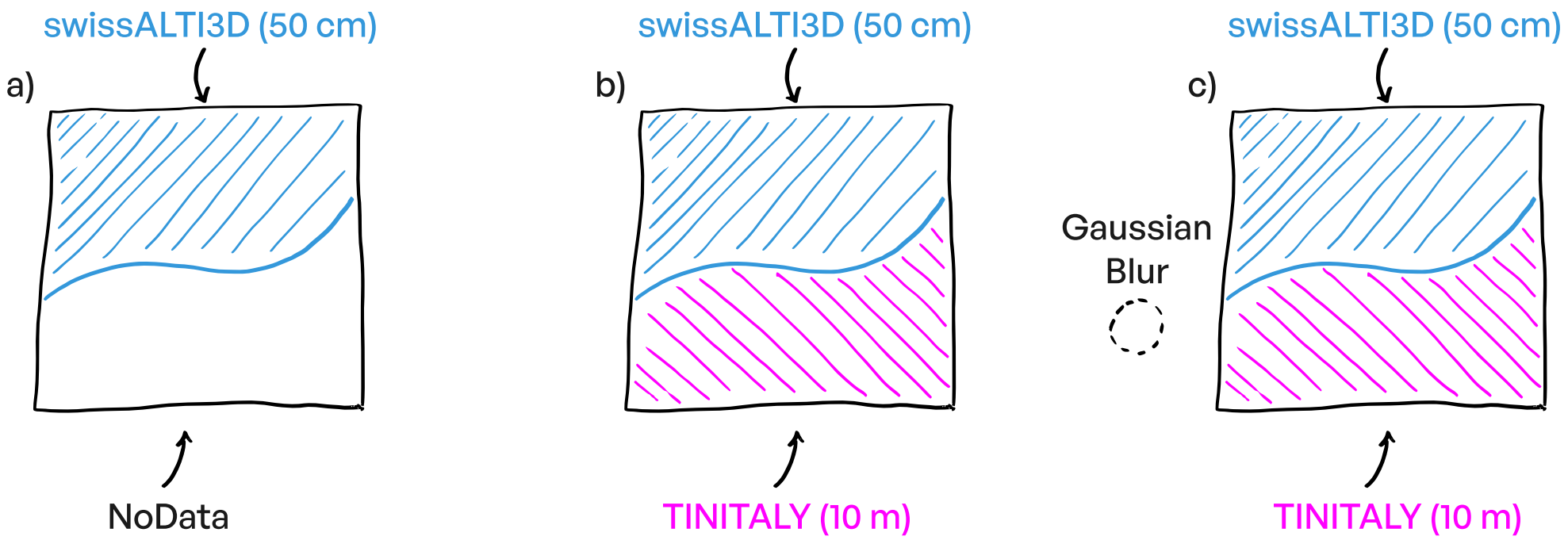
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Aggregation

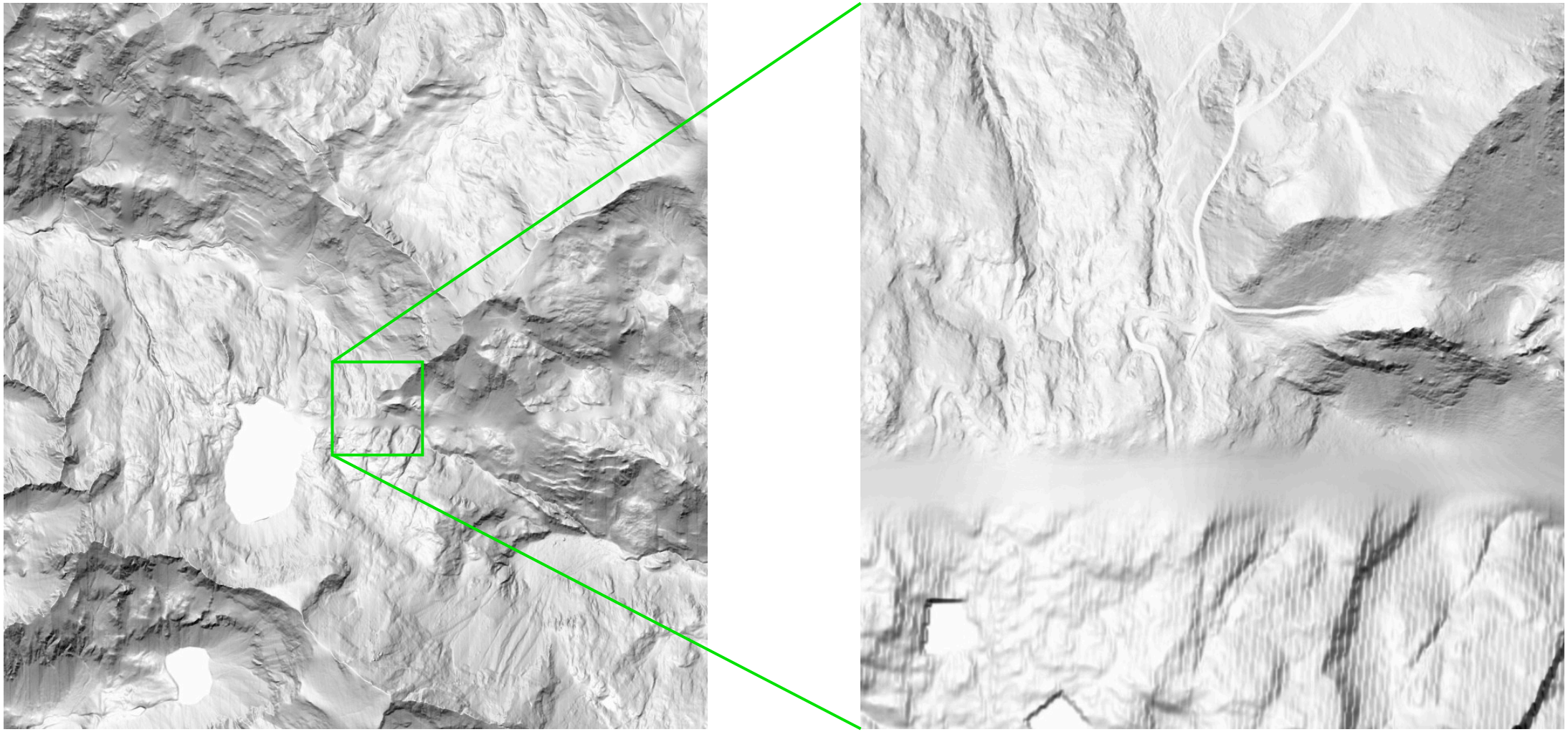


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Aggregation



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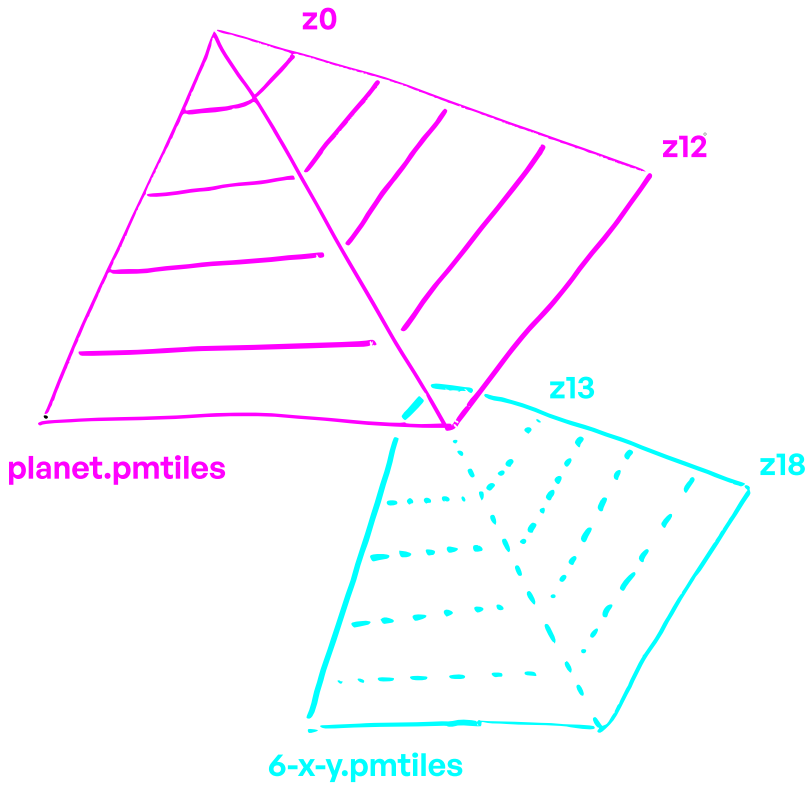
zoom 13.5

zoom 16.5

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Output

- 11.8 TiB elevation tiles
- 458 PMTiles archives
- planet.pmtiles: global, zoom 0-12
- 6-x-y.pmtiles: local, zoom 13-var
- mapterhorn.com/data-access



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Usage

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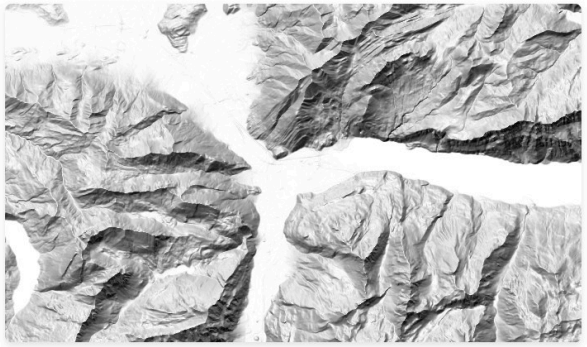
Viewer Data Access Coverage Attribution **Examples**

mapterhorn.com/examples

Examples

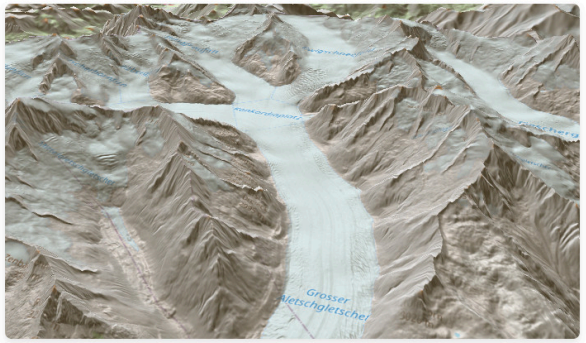
Hillshade

Add hillshading to a map.



Terrain

Add 3D terrain and hillshading to an OpenStreetMap raster map.



Contour

Add contour lines to a map.



PMTiles

Use Mapterhorn with PMTiles and addProtocol.



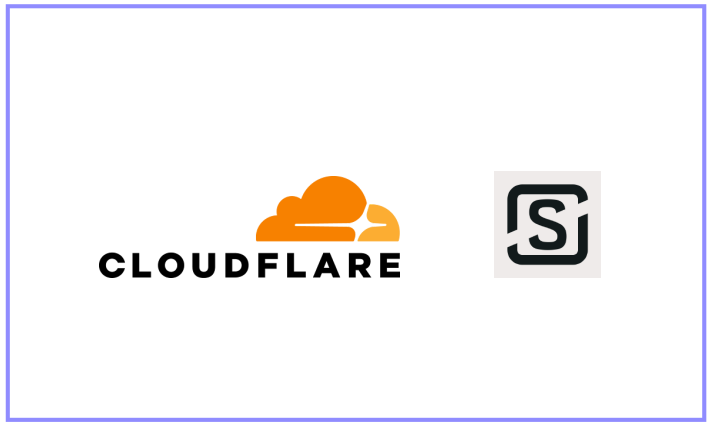
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Sponsors

opencollective.com/mapterhorn



Research & Development



Web Hosting

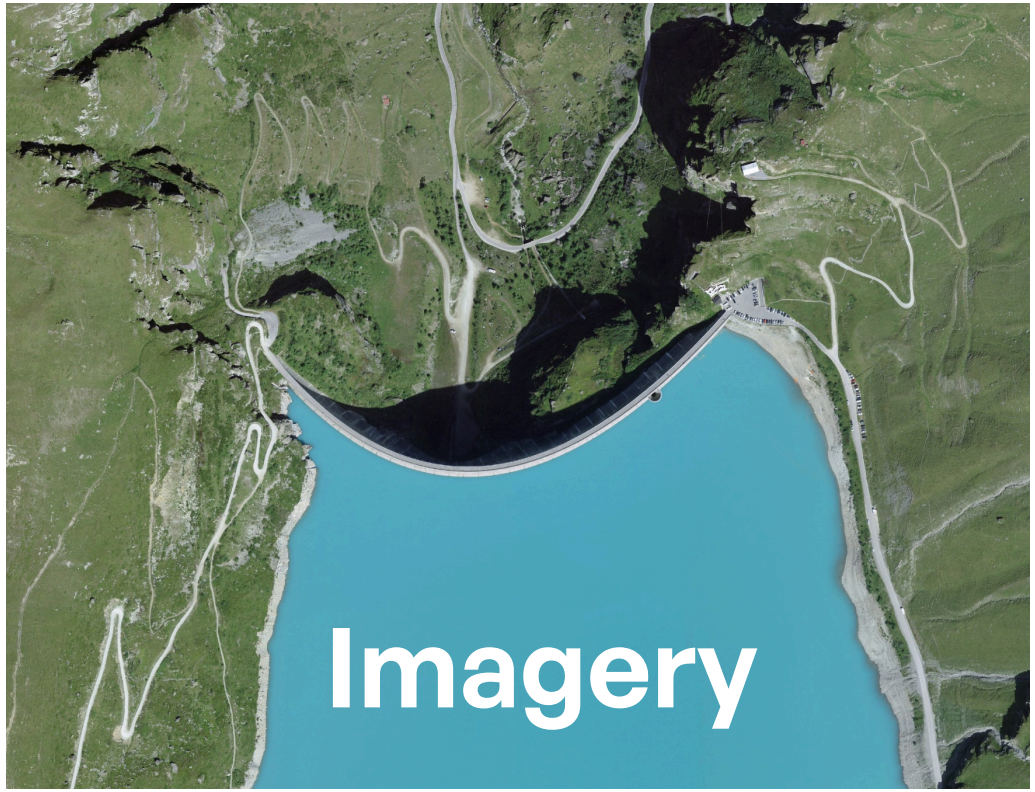


Maintenance

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Mapterhorn Imagery

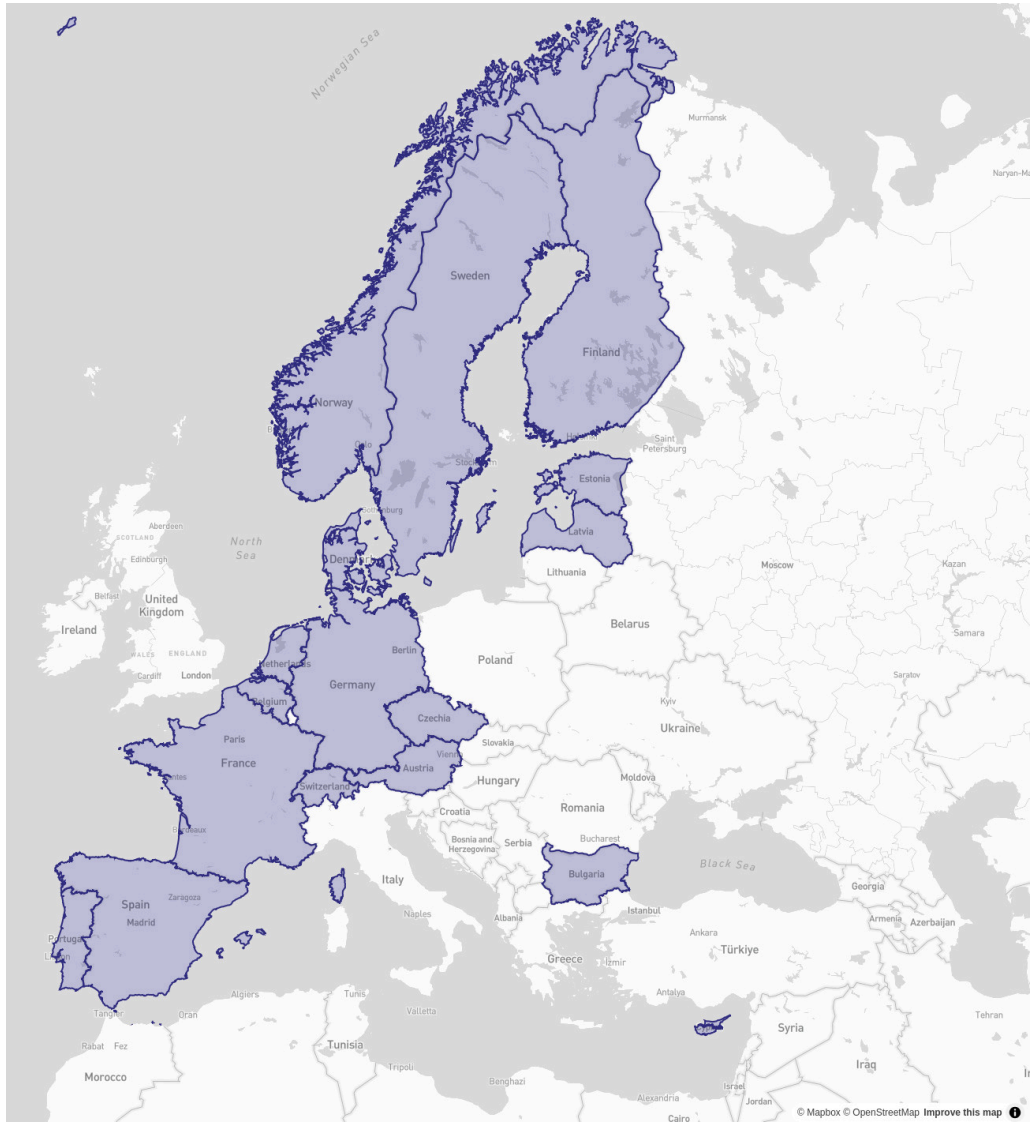
- Extend terrain pipeline to aerial imagery
- Alternative to VersaTiles Imagery
- Same producers, "same" format
- Updates more frequently
- Storage: 100+ TiB
- Prepare for Petabytes...



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Initial Imagery Coverage

- Global 10 m coverage Sentinel-2
- High-resolution: 18 countries and regions
- Focus first on Europe, then expand to JP, NZ, US, CA, etc...
- First release still in 2026



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